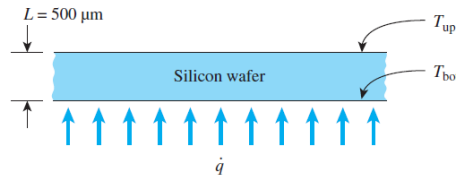


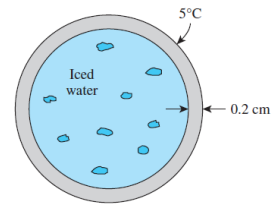
Heat Transfer: Practice Problems

Conduction

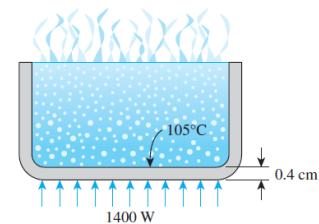
1. Silicon wafer is susceptible to warping when the wafer is subjected to temperature difference across its thickness. Thus, steps needed to be taken to prevent the temperature gradient across the wafer thickness from getting large. As an engineer in a semiconductor company, your task is to determine the maximum allowable heat flux on the bottom surface of the wafer, while maintaining the upper surface temperature at 27°C. To prevent the wafer from warping, the temperature difference across its thickness of 500 μm cannot exceed 1°C.



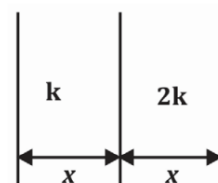
2. A hollow spherical iron container with outer diameter 20 cm and thickness 0.2 cm is filled with iced water at 0°C. If the outer surface temperature is 5°C, determine the approximate rate of heat loss from the sphere, in kW, and the rate at which ice melts in the container. The heat of fusion of water is 333.7 kJ/kg



3. An aluminum pan whose thermal conductivity is 237 W/m·K has a flat bottom with diameter 15 cm and thickness 0.4 cm. Heat is transferred steadily to boiling water in the pan through its bottom at a rate of 1400 W. If the inner surface of the bottom of the pan is at 105°C, determine the temperature of the outer surface of the bottom of the pan.

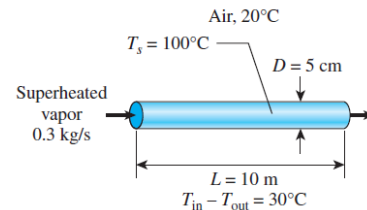


4. Establish a relation for the time taken to form a layer of ice on the surface of a pond. How much time it will take for a layer of ice of thickness 20cm to increase by 1mm on the surface of a pond when the temperature of surroundings is -20°C ? Thermal conductivity of ice = 7.5 kJ/m.hr.°C Latent heat of ice = 335 kJ/kg Density of ice at 0°C = 1000 kg / m
5. Derive an expression for temperature distribution and conduction heat flow in a circular conical rod with diameter at any section given by $D=ax$ where x is the distance measured from the apex of the cone and 'a' is a certain numerical constant. It may be presumed that there is no internal heat generation, steady state conditions prevail and the lateral surface is well insulated. Further proceed to obtain the numerical value for heat flow rate if the smaller end located at $x_1=60\text{mm}$ has a temperature 650K and the larger end at $x_2=300\text{mm}$ has a temperature of 450K. The parameter 'a' equals 0.20 and the material of the conical rod has an average thermal conductivity $k=3.45\text{ W/m}^{\circ}\text{C}$.
6. Two plates of equal thickness (t) and cross-sectional area are joined together to form a composite as shown in the figure. If the thermal conductivities of the plates are k and $2k$, then the effective thermal conductivity of the composite is



Convection and lumped system

1. Air at 20°C with a convection heat transfer coefficient of 20 W/m²·K blows over a pond. The surface temperature of the pond is at 40°C. Determine the heat flux between the surface of the pond and the air.
2. In a power plant, pipes transporting superheated vapor are very common. Superheated vapor is flowing at a rate of 0.3 kg/s inside a pipe with 5 cm in diameter and 10 m in length. The pipe is located in a power plant at 20°C, and has a uniform pipe surface temperature of 100°C. If the temperature drops between the inlet and exit of the pipe is 30°C, and the specific heat of the vapor is 2190 J/kg·K, determine the heat transfer coefficient as a result of convection between the pipe surface and the surrounding.



3. The temperature of a gas stream is to be measured by a thermocouple whose junction can be approximated as a 1-mm-diameter sphere, as shown in Fig. 4–10. The properties of the junction are $k=35\text{ W/m}\cdot\text{K}$, $\rho=8500\text{ kg/m}^3$, and $C_p=320\text{ J/kg}\cdot\text{K}$, and the convection heat transfer coefficient between the junction and the gas is $h=210\text{ W/m}^2\cdot\text{K}$. Determine how long it will take for the thermocouple to read 99 percent of the initial temperature difference. Assume lumped system approximation is applicable
4. Consider a potato being baked in an oven that is maintained at a constant temperature. The temperature of the potato is observed to rise by 5°C during the first minute. Will the temperature rise during the second minute be less than, equal to, or more than 5°C? Why?
5. Consider two identical 4-kg pieces of roast beef. The first piece is baked as a whole, while the second is baked after being cut into two equal pieces in the same oven. Will there be any difference between the cooking times of the whole and cut roasts? Why?
6. Consider a sphere and a cylinder of equal volume made of copper. Both the sphere and the cylinder are initially at the same temperature and are exposed to convection in the same environment. Which do you think will cool faster, the cylinder or the sphere? Why?
7. Obtain a relation for the time required for a lumped system to reach the average temperature $0.5(T_i + T_\infty)$, where T_i is the initial temperature and T_∞ is the temperature of the environment.
8. A person is found dead at 5pm in a room whose temperature is 20°C. The temperature of the body is measured to be 25°C when found, and the heat transfer coefficient is estimated to be 8 W/m²·K. Modeling the body as a 30-cm-diameter, 1.70-m-long cylinder and using the lumped system analysis as a rough approximation, estimate the time of death of that person. Specific heat of human body = 4178 J/(kg·K).
9. Consider a snow-making machine. It projects droplets with diameter of 0.5 mm into air at -20°C. If the initial temperature of the droplet is 6°C, how long does it take to become ice? Thermal properties of water or ice at 0°C: density $\rho = 1000\text{ kg m}^3$, specific heat $C = 4217\text{ J/(kg}\cdot\text{K)}$ and the latent heat of ice melting into water is $l_{sf} = 333.7 \times 10^3\text{ J/kg}$. Assume the droplet temperature is uniform through the volume, and the convection coefficient is $h = 200\text{ W/m}^2\text{K}$.